

# Teaching Crystallographic Computing with Computational Notebooks

**Blaine Mooers, PhD**  
**blaine-mooers@ou.edu**

Department of Biochemistry & Physiology  
University of Oklahoma Health Sciences Center, Oklahoma City

IUCr 2026

2016 August 17, 3:00-3:30 PM, Room 224

Slides and notebooks:

`MooersLab/iucr2026-teaching-computational-crystallography`

# Outline

The Curriculum Challenge and One Approach

The Computational Notebook Platform

Six Notebooks

Benefits and Limitations of Computational Notebooks

Acknowledgements

# The Curriculum Challenge

- Limited opportunities for teaching crystallography.
- Even more limited opportunities for teaching computational crystallography.
- Intense competition for slots in first year graduate curricula.
- Graduate students arrive with limited computing experience.
- Limited to covering these topics in group meetings and day long workshops.

# Solution: embed vignettes in existing lectures

- Limit to a short vignette of several notebook cells; do not have to run full notebook.
- No need for new course approvals.
- Minimal disruption to existing curricula.
- Reduces the abstract nature of crystallographic concepts.

# What Are Computational Notebooks?

- Computational narratives that combine executable code, output, and text.
- Jupyter notebooks are the most widely used format.
- Jupyter: Julia, Python, and R.
- Support polyglot computing via kernels.
- Students see cause and effect.
- Dopamine hits from instant gratification.
- Uses: code chunk development, data exploration, teaching, self-directed tutorials, and supplemental materials.



Kluyver et al. (2016) In F. Loizides & B. Schmidt (Eds.), Positioning and Power in Academic Publishing: Players, Agents and Agendas (pp. 8790). IOS Press.

# Platforms for Jupyter

- **Jupyter Lab:** (+) recommended, (+) IDE-like.
- Jupyter Notebook: (+) Classic and (+) modern variants.
- nteract.app: (+) drag-and-drop install, (-) no extensions.
- Google Colab: (+) no local install problems, (-) time-consuming virtual machine setup.

# Six demonstration notebooks

The notebooks are coded to run in Google Colab and Jupyter. Corresponding Python scripts are included.

- Demo 1: Data reduction with DIALS and Gemmi. Downloads images from Zenodo
- **Demo 2: Data reduction with XDS.**
- **Demo 3: MR with Phaser**
- **Demo 4: Refinement with Phenix**
- Demo 5: Advanced data analysis with reciprocalspaceShip
- Demo 6: Code reuse.

*If the software has a Python API or binaries that can be run in the terminal, you can run them in Jupyter.*

# DIALS in Jupyter

DIALS: Diffraction Integration for Advanced Light Sources

## Topics covered:

- Importing diffraction images from Zenodo
- Spot finding and indexing
- Refinement and integration
- Visualizing reciprocal space



DIALS: Winter et al. (2018) Acta Cryst. Sect. D 74, 86-97; 6YNQ: Gunther et al. (2021) Science 372, 642-646.

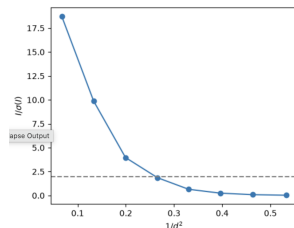


# XDS via Subprocess Calls

XDS: X-ray Detector Software

## Topics covered:

- Generate XDS.INP file.
- Run XDS from notebook cells.
- Parse CORRECT.LP for statistics.
- Compare results with DIALS.



Kabsch (2010) Acta Cryst. Sect. D 66, 125-132.

# Molecular Replacement Workflow

## Topics covered:

- Preparing search models with CHAINSAW or Gemmi.
- Running PHASER.
- Evaluating MR solutions.
- Initial refinement cycles.



McCoy, A. J. et al. (2007). J. Appl. Cryst. 40, 658-674.

# Refinement Workflow

## Topics covered:

- Setting up phenix.refine.
- R-free flag assignment.
- Monitoring convergence.
- Analyzing geometry with MolProbity.
- Iterative model improvement.



Phenix: Liebschner, D. et al. (2019). Acta Cryst. D75, 861-877; Molprobity: Williams, C. J. et al. (2018). Protein Sci. 27, 293-315

# reciprocalspaceship and Gemmi

## Topics covered:

- DataFrames for reflection data (reciprocalspaceship).
- Fast structure parsing with Gemmi.
- Custom data transformations.
- Integration with pandas and NumPy.

reciprocalspaceship: Greisman et al. (2021). J. Appl. Cryst. 54, 1521-1529; Gemmi: Wojdyr (2022). J. Open Source Software 7, 4200.

# Reducing Errors with Snippets

## **Topics covered:**

- Code snippet libraries as software engineering tools.
- GhostText for tab triggers in notebooks.
- Advantages over AI-generated code for beginners.
- Building personal snippet collections.

# Advantages of Computational Notebooks

- Cause-and-effect visible for students.
- Workflow documented in one file.
- Easy to share via GitHub or Codeberg repositories\*.
- Integration of narrative explanations with code.
- Foundation for reproducible research practices.

BEWARE: May not be able to share course materials publicly.

# Limitations

- Heavy-duty data processing still needs established pipelines.
- Must prepare and select the correct jupyter kernel.
- Diffraction images should be downloaded in advance.
- Notebook pipelines slower than established GUIs.
- Software installs and kernel creation can frustrate.
- Students struggle with debugging Python errors.
- Not all crystallographic software has Python APIs.
- Instructor must develop notebooks – lengthens lecture preparation time.

# Teaching Reproducible Research

- Notebooks naturally document computational provenance.
- Version control with Git teaches good data practices.
- Links to deposited data (PDB, SBGrid) reinforce FAIR principles.
- Students learn to value documentation as part of science.
- Reproducibility concerns are woven into every lesson.

**FAIR:** Findable, Accessible, Interoperable, Reusable



# Acknowledgements

## **Funding:**

- NIH: R01 CA242845, R01 AI088011
- NIH: P20 GM103640, P30 CA225520, P30 AG050911-07S

## **Thanks to:**

- All of the developers of crystallographic software.
- Christopher O'Grady, SLAC.
- Graduate students who provided feedback.
- Oklahoma Data Science Workshop community.

# Registration Still Open



The banner features a background with a blue-to-red gradient and a geometric pattern of overlapping squares and lines. The text is white and yellow, providing high contrast against the background.

**Bay Area Light Sources Joint Users' Meeting**  
September 20-25, 2026  
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<https://web.cvent.com/event/313320d8-d854-4c11-8501-76e24fcae513/summary>

# Submissions Open to Presenters

special issue submission

## Bay Area Light Sources Joint Users' Meeting 2026

Edited by Blaine Mooers and Chitra Rajendran



Following the commissioning of a special issue from last year's *Stanford Synchrotron Radiation Lightsource (SSRL) and Linac Coherent Light Source (LCLS) Joint Users' Meeting* (in progress at [http://journals.iucr.org/special\\_issues/2026/ssrl-lcls/](http://journals.iucr.org/special_issues/2026/ssrl-lcls/)), IUCr Journals are pleased to invite speakers at the *Bay Area Light Sources Joint Users' Meeting* to contribute to a new special issue.

Workshop speakers, poster presenters and workshop organizers are all welcome to submit articles. Contributions need not be identical to the presentation made, and can include research papers, methods papers and topical reviews. Please note that meeting reports or short conference proceedings contributions will not be suitable.

Submissions to the special issue will be considered for publication in the following IUCr Journals:

- *Acta Crystallographica Section D: Structural Biology*
- *Journal of Applied Crystallography*
- *Journal of Synchrotron Radiation*
- *Acta Crystallographica Section A: Foundations and Advances*
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